Equations

Nernst Equation

$$E_{ion} = \frac{RT}{zF} ln \frac{[ion]_o}{[ion]_i}$$

for T in Kelvin, R = 8.314 *J/K*mol*, and F = 96, 485 *C/mol*

ln = natural log, base e (~2.718)

GHK Equation

$$Vm = \frac{RT}{F} ln \frac{P_k[K]_o + P_{Na}[Na]_o + P_{Cl}[Cl]_i}{P_k[K]_i + P_{Na}[Na]_i + P_{Cl}[Cl]_o}$$

for T in Kelvin, R = 8.314 *J/K*mol*, and F = 96, 485 *C/mol*

Driving Force

$$Driving\ Force = V_{membrane} - E_{ion}$$

Ohms Law

$$Current\ (I) = \frac{Voltage\ (V)}{Resistance\ (R)}$$

Current in Amps (*A*), Voltage in Volts (*V*), Resistance in Ohms (*Ω*)

Length Constant

$$Length\ Constant\ (\lambda) = \sqrt{\frac{R_{membrane}}{R_{axial}}}$$

$R_{membrane}$ in *Ω*cm*, R_{axial} in *Ω/cm*

Time Constant

$$Time\ Constant\ (\tau) = R_{membrane} C_{membrane}$$

$R_{membrane}$ in Ω*cm, $C_{membrane}$ in F